



AI weather forecast

A study utilised AI to pinpoint areas where weather changes rapidly and focus their resources there. <https://digit.in/sep19-19>



First photo from Chandrayaan 2

ISRO tweeted the first photo from Chandrayaan 2 showing the Apollo crater and the Mare Orientale basin. <https://digit.in/sep19-20>

supermassive black hole, increased the amount of material emanating from the astrophysical jets of the black holes. These are highly ionised matter travelling at relativistic speeds (near the speed of light) from the “poles” of the black holes, or the axis of rotation. These jets push out the surrounding gas and material into the halo, from where they fall towards the galaxy core again, restarting the entire process. The supermassive black hole is acting like a pump in the garden fountain, the pool of water being the halo of the galaxy. Grant Tremblay, an astrophysicist who worked on the observations says, “The supermassive black hole at the center of this giant galaxy acts like a mechanical ‘pump’ in a water fountain. This is one of the first systems in which we find clear evidence for both cold molecular gas inflow toward the black hole and outflow or uplift from the jets that the black hole launches.” The fountain has as much matter as three billion Suns. Astronomers speculate that such fountains may exist in other galaxies as well. NGC 5775 is one such galaxy, which is advantageous to astronomers as we can mostly see the galaxy from the side, or the edge of the spiral. The plane of the galaxy is perpendicular to us. Due to this orientation, the regions forming the halo above and below the plane of the galaxy is clearly visible to astronomers. There is a clear halo around NGC 5775, and it is believed to have originated from the large number of new stars being formed there. The galactic fountain mechanism is suspected to exist within the Milky Way as well, but our only indications are from the observations of the galactic halo. David Henley from the University of Georgia explains the possibility of a galactic fountain within the Milky Way, “Although we cannot directly observe the hot gas rising out of the disc, it has long been suspected that galactic fountains are responsible for the hot gas observed in the Milky Way’s halo. We have taken spectra of the X-ray-emitting hot gas

in the galactic halo and compared them to detailed predictions coming from different models. The galactic fountain scenario turned out to be the one that best describes our data.”

The material towards the center of the galaxy is cold, while the material just ejected out into the halo is hot. This hot material condenses or clumps up, and then falls back in towards the galaxy. The movement of matter within and around the galaxy is similar to the water cycle on Earth. The infalling material is known as galactic rain. While such galactic

on the galaxy, fueling star formation and feeding its supermassive black hole.” The astronomers identified three distinct clouds or clumps of matter, falling towards the black hole. Each of these clouds had as much matter as a million Suns, spread out over tens of light years. These clouds could be observed because they temporarily obscured the light that normally would reach the Earth from Abell 2597. In effect, these clouds were casting a billion light year long shadow. While the astronomers only observed three of these clouds, they



Credit: ESA

An illustration of how astronomers think the galactic fountains in the Milky Way work. The material from supernovae are ejected into the surrounding halo, from where they fall back into the galaxy when they cool. The fountains shown here are a rough indication, and it is unknown how many are there in the Milky Way. The yellow dot indicates the position of the Sun.

rain had been theoretically predicted since the 70s, it has only been directly observed in 2016, also around the Abell 2597 cluster. The observations shed light on how black holes feed on matter, and show that the diet of black holes consists of frenzied feeding events, apart from the steady inflow of matter from their accretion discs. An astronomer from Yale University noted, “It’s exciting to think we might actually be observing this galaxy-spanning rainstorm feeding a black hole whose mass is about 300 million times that of the Sun. This very, very hot gas can quickly cool, condense, and precipitate in much the same way that warm, humid air in Earth’s atmosphere can spawn rain clouds and precipitation. The newly condensed clouds then rain in

speculate that there could be over a thousand of them in the vicinity of Abell 2597.

When the black hole gorges on these clouds, the output and energy of astrophysical jets increase, heating the cool gas surrounding the black hole. This heat prevents the formation of more clouds, and also limit the amount of material available for the formation of new stars. One of the astronomers on the team who made the observations, Mark Voit from Michigan State University explains, “One of the fundamental questions in astronomy is what regulates the growth of galaxies. We now have more evidence that this ‘rain’ slows down star formation in galaxies by feeding the huge black holes at their centers.”